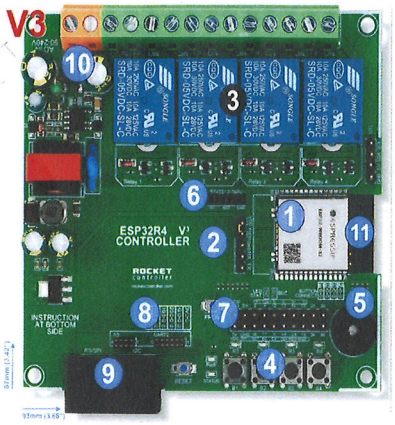
ROCKET controller **rocketcontroller.com**

visit for tech support, firmware updates, user guides, smarthome integrations, projects

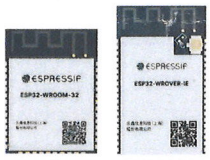


ESP32R4 V3 CONTROLLER

ESP32 MCU, 2.4GHz Wi-Fi & Bluetooth LE 4.2

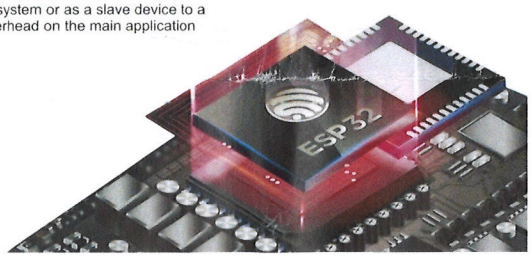
1. Flash memory 4MB and extend to 16MB + 8MB PSRAM
2. UART programming header with PROG and RESET buttons
3. 4 Contact Relays 10A 250VAC/30VDC with LED indicate
4. 4 push Buttons
5. Sound Buzzer
6. Socket for RF433/315 receiver module
7. Header for connecting LCD TFT touch display
8. Socket for embedded communication modules
9. RJ-12 socket for expansion modules
10. AC 90~240V or DC 7~36V power IN for variation
11. Imbedded antenna or external antenna 8dBi

Powerful with ESP32 microcontroller, 2.4 GHz Wi-Fi and Bluetooth LE 4.2, ESP32R4 can perform as a complete standalone system or as a slave device to a main controller, reducing communication stack overhead on the main application processor.



4MB FLASH

16MB FLASH
8 MB PSRAM



Characteristics

Dimension	
PCB	93x87mm
Enclosure	115x90x45mm

Connections	
Button 1	GPIO34
Button 2	GPIO35
Button 3	GPIO36
Button 4	GPIO39
Relay 1	GPIO25
Relay 2	GPIO26
Relay 3	GPIO33
Relay 4	GPIO32

Header UART	
3.3V	Power IN
ESP VIN	ESP32 power IN (3.3V)
GND	GND
RXD	GPIO3
TXD	GPIO1

Header IO	
3.3V	DC IN 3.3V
5V	DC IN 5.0V
EN	EN (button RST)
GND	GND
IO2	GPIO2
IO4	GPIO4
IO5	GPIO5 VSPI CS0
IO12	GPIO12 HSPI MISO
IO13	GPIO13 HSPI MOSI
IO14	GPIO14 HSPI CLK
IO15	GPIO15 HSPI CS0
IO16	GPIO16 UART2 RXD
IO17	GPIO17 UART2 TXD
IO18	GPIO18 VSPI CLK
IO19	GPIO19 VSPI MISO
IO23	GPIO23 VSPI MOSI

Header LCD/I2C	
GND	GND
5V	DC IN 5.0V

SDA	GPIO21 I2C SDA
SCL	GPIO22 I2C SCL

Header RF433	
5V	DC IN 5.0V
GND	GND
IO27	GPIO27

Wi-Fi/Bluetooth Controller ESP32-D0WDQ6	
Wi-Fi series/standard	802.11 b/g/n • 802.11 n (2.4 GHz), up to 150 Mbps
Bluetooth	Bluetooth v4.2 BR/EDR and BLE specifications
Output Power	+12 dBm transmitting power
Receiving sensitivity	NZIF receiver with –94 dBm BLE sensitivity
CPU	Xtensa® single-/dual-core 32-bit LX6 microprocessor(s), up to 600 MIPS

Flash memory	
Type/architecture	NOR FLASH
Memory capacity	32Mb (4M x 8)
Memory interface type	SPI - Dual/Quad I/O
Operating Voltage	2.7V ~ 3.6V

System Buttons	
RST	EN ESP32 IC (RESET)
PROG	GPIO0 ESP32 IC (PROGRAMMING/FLASH MODE)

Relays	
Power limit	10A 250VAC/30VDC
Limit time	100 000 cycles

AC-IN power supply	
AC IN	85~265V
DC OUT	5V 650mA
Circuit	Isolation transformer
Power consumption	88% efficiency
Protection	DC out short circuit protection

DC-IN power supply	
DC IN	DC 7~36V
DC OUT	5V 3A, 3.3V 0.8A
Protection	DC IN polarity & short circuit protection